

Predicting α -helix secondary structure

A two-state Hidden Markov Model fit with amortized BayesFlow inference

Group [number]

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Roadmap

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The question Predict α -helix vs other from sequence

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Forward–Backward Exact per-residue posterior as training targets

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BayesFlow Amortized neural posterior via a sliding window

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Insulin & real evaluation Held-out real proteins vs ground truth

5

Comparison & results BayesFlow vs exact Forward–Backward

From amino-acid sequence to α -helix

- Proteins fold; the local shape is the secondary structure (helix / sheet / coil).
- We predict one thing: is each residue an α -helix, or 'other'?
- Solving 3-D structure needs X-ray / NMR — expensive; we want it from sequence.
- Our statistical model: a two-state Hidden Markov Model.

α -helix = state “H”

“other” = E B G I T S C

Mapping decision

DSSP labels each residue with 8 states. We keep only H as helix (strict α -helix); the other 7 $\rightarrow$ other.

The generative model: a two-state HMM

- The chain always starts in “other”.
- Transitions: helix→helix 0.90, other→helix 0.05.
- So helices form ~10-residue runs, and are started rarely.
- Realistic: real data gives 0.912 and 0.041 (see next slide).

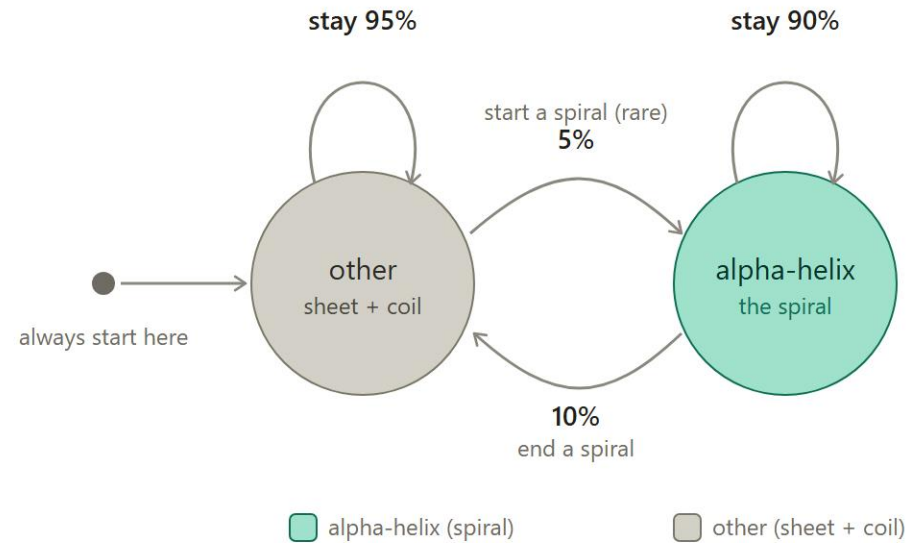


Fig 1. Two-state HMM: start rule and transition probabilities.

Emissions — and are they realistic?

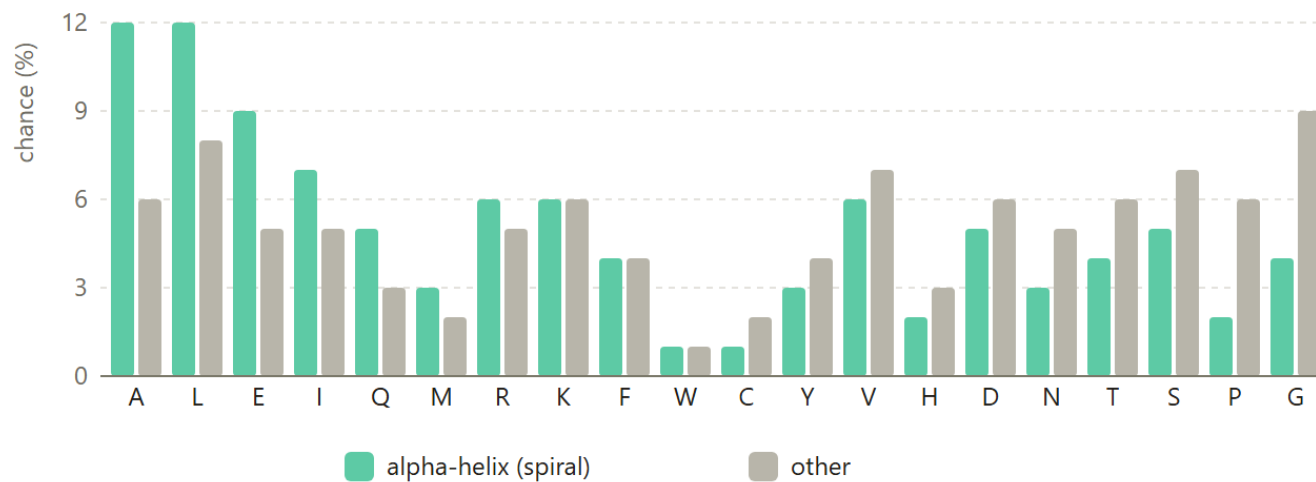


Fig 2. Per-state amino-acid emission probabilities (α -helix vs other).

Model checks out against real data: $P(\text{helix} \rightarrow \text{helix})$ 0.912 vs 0.90 · $P(\text{other} \rightarrow \text{helix})$ 0.041 vs 0.05 · mean helix run 11.4 vs 10 · 100% of real chains start in “other”.

Notebooks 1–2: simulate, then get exact targets

- Simulate 100,000 chains; lengths sampled from real PISCES (median 208).
- Run Forward–Backward (hmmlearn) → exact $P(\text{helix})$ per residue = training targets.
- 24.3M residues — the exact Bayesian “answer key”.
- Checks: mean P 0.324; residue-0 = 0; recovers hidden states at the Bayes ceiling (AUC 0.79).

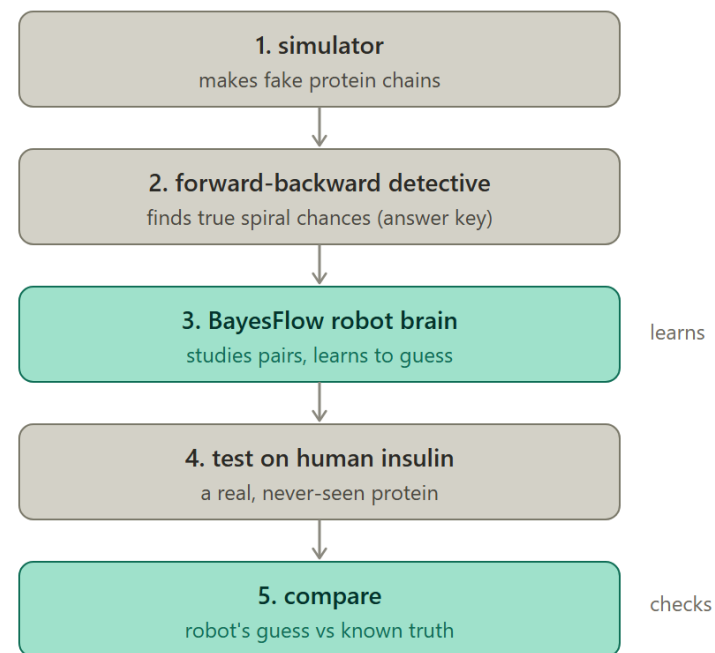


Fig 3. Full pipeline: simulator → Forward–Backward → BayesFlow → evaluation.

Turning a variable-length chain into a fixed input

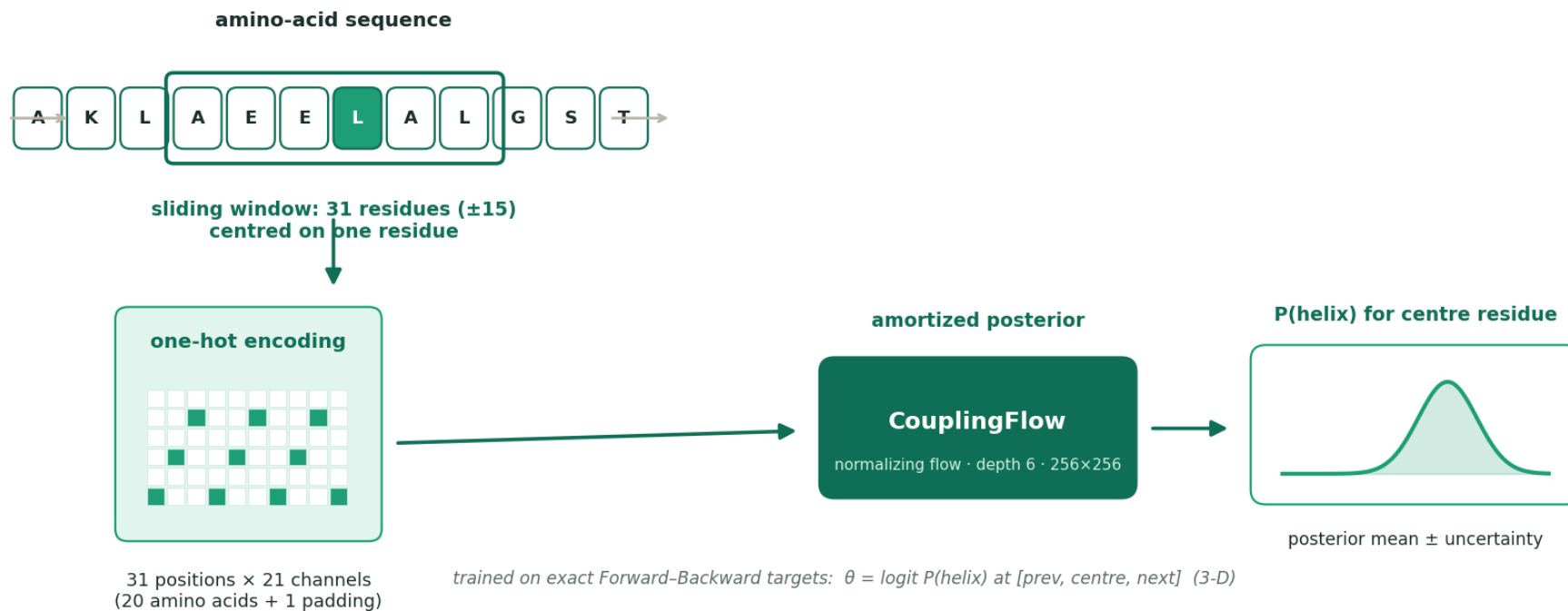


Fig 4. Sliding 31-residue window $\rightarrow$ one-hot encoding $\rightarrow$ CouplingFlow posterior.

- A flow needs a fixed-size target $\rightarrow$ predict one residue from a fixed 31-residue window (± 15), slid along the chain.
- Encode the window as one-hot: 31 $\times$ 21 channels = 20 amino acids + 1 padding (for chain ends).
- The window is only local $\rightarrow$ the answer stays uncertain $\rightarrow$ a genuine posterior (mean $\pm$ std), not a point estimate.

How we trained it

Component	Choice · why
Inference network	CouplingFlow, depth 6, 256×256 subnets
Target θ (3-D)	logit $P(\text{helix})$ at [prev, centre, next]
Conditions	flattened $31 \times 21 = 651$ one-hot window
Training data	300,000 windows from 15,000 chains
Optimizer / epochs	Adam · 30 epochs · batch 512

No leakage (verified)

Train = front block [0, 15000); validation = tail block [99500, 100000). Disjoint indices AND 0 shared sequences — asserted at runtime.

Why a window?

One trained model does inference for a chain of ANY length — just slide the window. Amortized: no re-fitting per protein.

Held-out real protein: human insulin (1A7F)

- Insulin + its true DSSP labels pulled straight from the dataset.
- Model trained ONLY on simulated data → insulin is genuinely unseen (no leakage).
- $P(\text{helix})$ rises inside the annotated helix on both chains.
- AUC: chain A 0.97, chain B 0.98 — BayesFlow = exact FB.

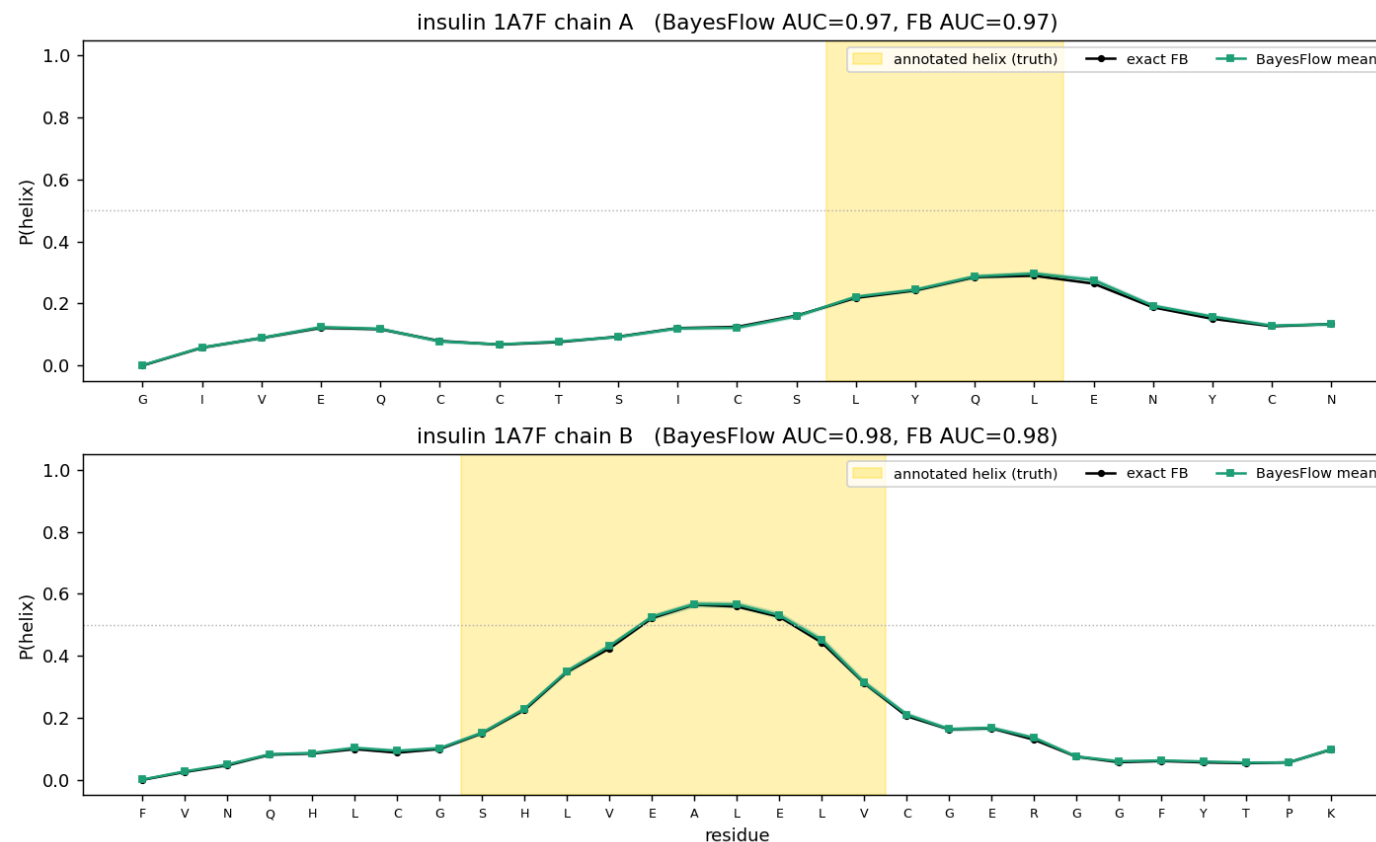


Fig 5. Insulin $P(\text{helix})$: BayesFlow (green) vs exact FB (black); gold band = true helix.

All real proteins (PISCES)

- Scored all 8,994 standard-AA chains (2.2M residues).
- Per-chain AUC 0.754; BayesFlow = exact FB.
- A simple HMM ranks helices meaningfully across the messy real proteome.
- AUC > accuracy: the HMM isn't calibrated to reality → use ranking.

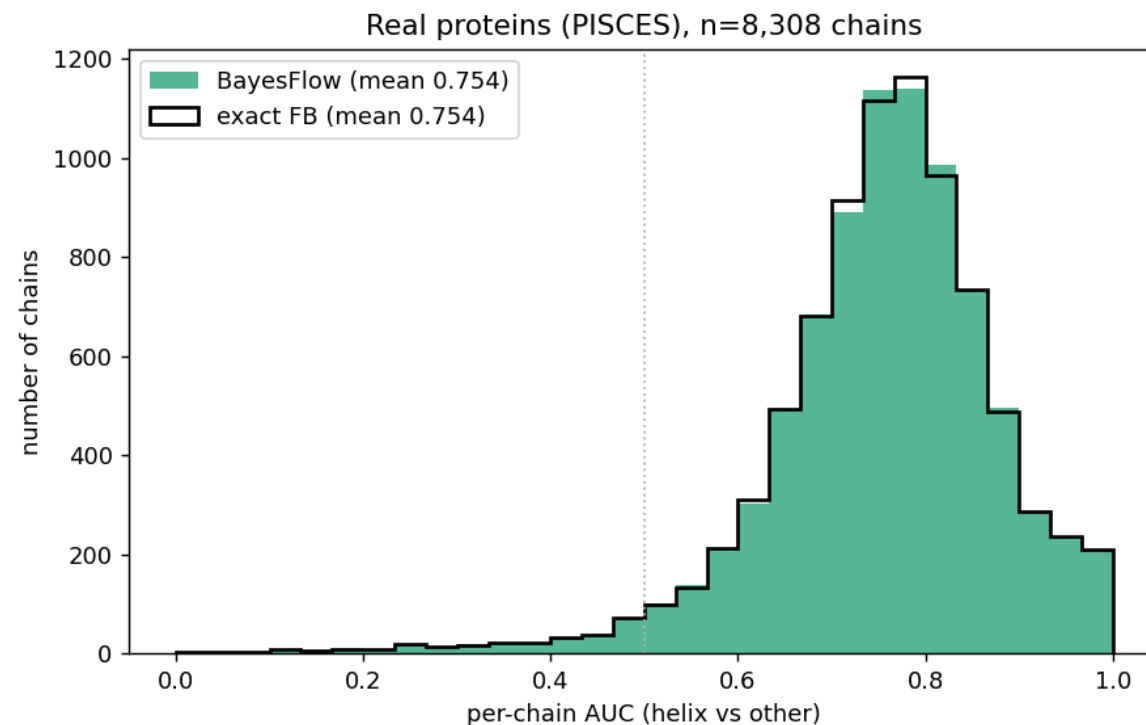


Fig 6. Per-chain AUC over 8,308 real chains (those with both classes).

Result 1: the neural posterior reproduces exact inference

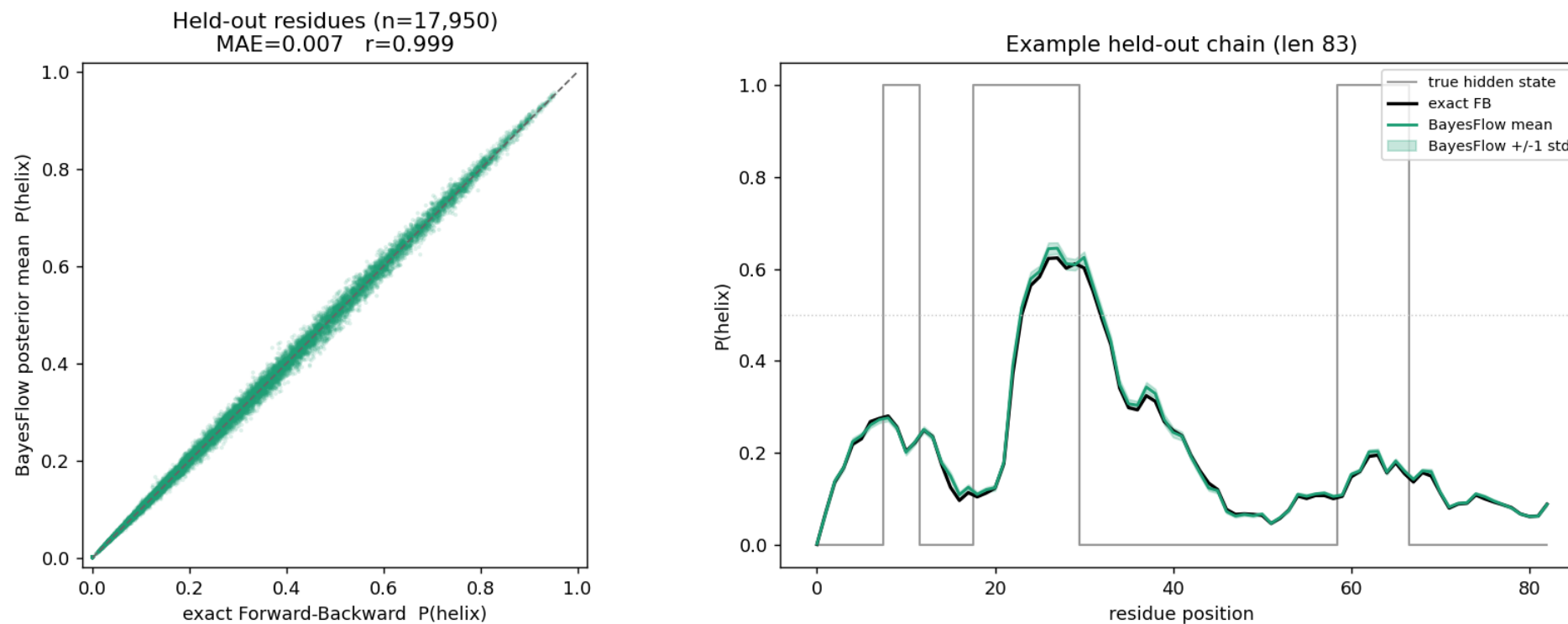


Fig 7. Left: BayesFlow mean vs exact FB ($r=0.999$). Right: an example chain with the ± 1 std band.

Held-out simulated chains: correlation 0.999, MAE 0.007. The fast neural posterior is interchangeable with slow HMM Forward–Backward.

Result 2: head-to-head across every setting

BayesFlow vs exact Forward-Backward, scored against ground truth

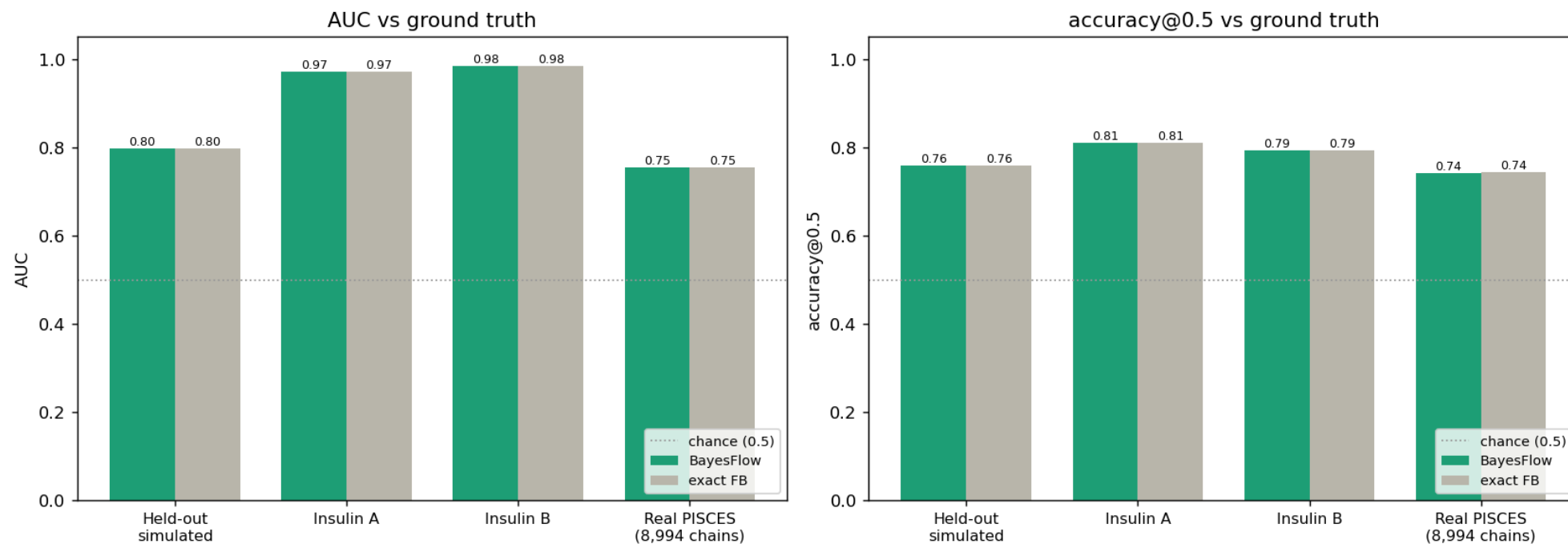


Fig 8. AUC and accuracy@0.5: BayesFlow vs exact FB across all four settings.

BayesFlow and exact FB are identical everywhere. High on insulin's textbook helices (0.97–0.98); ~0.75 across the real proteome.

Results at a glance

Setting	AUC BayesFlow	AUC exact FB	Acc@0.5 BayesFlow	Acc@0.5 exact FB
Held-out simulated	0.798	0.798	0.760	0.760
Insulin A (real, 1A7F)	0.971	0.971	0.810	0.810
Insulin B (real, 1A7F)	0.984	0.984	0.793	0.793
Real PISCES (8,994 chains)	0.754	0.754	0.743	0.743

Table 1. AUC and accuracy@0.5 — BayesFlow vs exact Forward–Backward, all settings.

- BayesFlow $\approx$ exact FB in every setting (AUC identical to 3 decimals).
- Both approach — never beat — the exact Bayesian answer (the AUC ceiling).
- Accuracy < AUC on real data: a calibration gap, expected for a fixed HMM.

TL;DR — take-home message

- α -helix modeled as a 2-state HMM whose transition probabilities match real proteins (0.912 vs 0.90).
- Trained a BayesFlow amortized posterior on exact Forward–Backward targets via a sliding 31-residue window.
- It reproduces exact FB almost perfectly ($r = 0.999$) — instant inference, no HMM at test time.
- On unseen real proteins it ranks helices well (insulin 0.97–0.98; PISCES 0.75), bounded by the model's Bayes ceiling.

α

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